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RESEARCH ARTICLE

Explainable AI Framework Using XGBoost With SHAP and LIME for Multi-Scale Household Energy Forecasting

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ABSTRACT Forecasting energy consumption is critical for maximizing resource utilization, lowering operational costs, and increasing sustainability. Traditional machine learning models often function as black boxes, which complicates the understanding of decision-making processes. This research utilizes Explainable Artificial Intelligence (XAI) methodologies to improve the interpretability of Extreme Gradient Boosting (XGBoost) to effectively predict energy consumption patterns. Shapley additive explanations (SHAP) is applied to reveal the significance of both global and local features, thereby clarifying how various factors of energy consumption influence the forecasts. Furthermore, Local Interpretable Model-Agnostic Explanations (LIME) contribute to model interpretability by providing a detailed examination of individual predictions. The proposed model identifies key determinants affecting energy usage through Permutation Feature Importance (PFI), while Partial Dependence Plots (PDPs) illustrate the relationships between input variables and the model's predictions. Additionally, Holt-Winters Exponential Smoothing is incorporated as a time-series decomposition method to investigate both short and long-term trends to predict energy consumption across various time intervals: hourly, daily, weekly, monthly, quarterly, as well as at the appliance level. This approach achieves a high R^2 value of 0.9999 and low RMSE values of 0.00344. Comparisons with baseline models demonstrate the superior performance of the proposed method. By integrating Holt-Winters + XGBoost with XAI (HW+XGB+XAI) techniques, this study enables AI-driven energy management and home energy optimization, ensuring both interpretability and trust.

INDEX TERMS Household energy consumption forecasting, explainable AI, XGBoost, Holt-Winters exponential smoothing, SHAP, LIME, permutation feature importance, energy management systems.

I. INTRODUCTION

Forecasting energy consumption is essential in contemporary energy management systems, facilitating cost efficiency, and promoting sustainable development, and ensuring optimal resource distribution [1]. Traditional machine learning techniques, such as XGBoost, have demonstrated high accuracy in predicting energy usage patterns by analysing various factors, including voltage, current, and sub-metering information [2]. However, these models often function as black boxes, providing minimal insight into how predictions are generated. This opacity can hinder practical applications and diminish stakeholder trust in energy management. To address this challenge, XAI methods are employed to interpret model predictions, ensuring that all stakeholders can understand and validate the decision-making process [3].

As part of XAI, SHAP and LIME give simple, model-independent ways to clarify individual predictions, making it easier to connect complex machine learning models and human understanding [4]. SHAP details provide both

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global and local interpretability by elucidating individual predictions and the overall importance of features, while LIME offers insights specific to individual instances within the model's decision-making procedure [5]. Additionally, PDPs contribute to model transparency by demonstrating the relationships between input variables and their corresponding predictions. This study employs exponential smoothing for time series prediction alongside machine learning explainability to analyse energy usage trends over various time stamps [6]. By combining XGBoost with XAI techniques, this project improves the interpretability of AI-driven energy analytics, making it more suitable for practical applications such as smart grid management, industrial energy optimization, and residential power efficiency [7].

To improve energy efficiency in residential environments within smart cities, a machine learning-driven energy forecasting model is introduced in [8]. The research work presented in [9] highlights the importance of XAI techniques, such as SHAP, in identifying key factors that influence energy usage. The study in [10] explores the use of QLattice as an XAI framework for predicting energy demand, demonstrating superior prediction accuracy and interpretability compared to traditional machine learning approaches like Random Forest and XGBoost. Additionally, a model based on Light Gradient-Boosting Machine (LightGBM) for energy consumption prediction is developed in [11]. The XAI methods discussed in [12] to examine the impact of specific variables on household energy usage reveal that submetering, which indicates the consumption of water heaters and air conditioners, plays a crucial role in the predictions. Furthermore, [13] creates a dynamic way to combine demand response with short-term load forecasting within a residential energy management system.

To improve the accuracy of forecasting, research [14], [15] employs an ensemble model that integrates XGBoost and Artificial Neural Networks (ANN), illustrating its effectiveness in handling peak-hour demand shifts. The study referenced in [16] investigates the performance of gradient boosting techniques, including XGBoost, CatBoost, and LightGBM, in forecasting energy usage. To further refine prediction accuracy, the research incorporates time series analysis techniques such as phase-space plots and wavelet transforms. In [17] and [18] a comparative analysis of Random Forest and XGBoost is conducted to assess their efficiency in forecasting electrical energy usage. The results indicate that Random Forest provides more precise energy consumption predictions with the lowest root mean square error across residential, commercial, and industrial domains. A novel two-stage methodology in [19] that merges statistical feature extraction with deep learning to forecast energy demand. This method aims to tackle significant challenges in predictive modelling for energy providers by improving predictive precision while maintaining interpretability. Furthermore, it evaluates various prediction models to estimate household energy consumption using multiple assessment metrics, including MSE, MAE, RMSE and (R^2) [20]. Predictions are generated on previously unseen data, and the optimal model is identified through comparative analysis. To elucidate the predictions, two XAI frameworks are deployed: SHAP and LIME [21].

To successfully implement energy forecasting in practical situations, it is essential to not only focus on the accuracy of the model but also to guarantee interpretability, scalability, and real-time applicability [22]. Contemporary urban infrastructures, such as smart grids and home automation systems, require intelligent solutions that can adjust to changing energy patterns and provide actionable insights [23]. The incorporation of XAI within the predictive modeling framework tackles this issue by allowing both system developers and non-technical stakeholders to comprehend and trust the outputs of the model [24]. This level of transparency promotes improved decision-making and encourages the acceptance of AI-driven energy management systems. Energy consumption patterns are affected by numerous factors, including weather conditions, occupancy behavior, appliance usage, and socio-economic variables. Consequently, models must be adept at capturing both deterministic and stochastic elements within the data [25]. While XGBoost is proficient in modeling intricate non-linear relationships, it does not inherently provide interpretability. To remedy this, SHAP and LIME act as complementary tools SHAP delivers both global and local model insights, while LIME offers intuitive explanations based on individual instances. The combination of these tools creates a robust interpretability layer over the predictive framework [26].

Moreover, employing traditional time series models like Exponential Smoothing aids in recognizing recurring seasonal patterns and long-term consumption trends, which are frequently neglected by tree-based algorithms [27]. The hybrid approach introduced in this study integrating machine learning with classical forecasting methods ensures resilience across various time granularities, ranging from hourly to quarterly forecasts. This dual strategy allows the system to address both short-term and long-term energy management requirements. The research further underscores the significance of feature importance analysis, which facilitates the identification of crucial factors such as sub-metering values, voltage, and global-active power. These findings are essential for utilities and policy-makers who seek to execute targeted interventions for demand-side management. By prioritizing interpretability in conjunction with performance, the suggested system aids in the overarching objective of developing transparent, accountable, and sustainable AI systems within the energy sector.

The main Contributions of this research are summarized as follows:

- Developed a robust model that integrates Holt-Winters exponential smoothing with XGBoost, effectively combining time-series decomposition with machine learning to enhance energy consumption prediction accuracy.
- Extracted and utilized trend, and seasonal components from Holt-Winters exponential smoothing as input

features, enabling the XGBoost model to learn temporal patterns more effectively than using raw time-series alone.
- Applied SHAP and LIME to provide both global and local interpretability, providing transparent insights into the model's decision-making.
- PFI and PDPs are applied to identify and visualize the most influential factors affecting energy usage patterns.
- Validated the proposed Holt-Winters + XGBoost model using key metrics such as MAE, MSE, RMSE, and R^2. The model was benchmarked against five widely adopted machine learning algorithms: Linear Regression, LASSO, KNN, Random Forest, and standalone XGBoost, as well as two established time-series forecasting methods: ARIMA and Prophet. The results consistently indicate that the proposed hybrid model outperforms all comparative approaches across multiple evaluation criteria.

The structure of this paper is organized as follows: Section II presents the framework for energy consumption prediction adopted in this study. Section III details the experimental setup and implementation details. Section IV reports the energy consumption forecasting results, followed by Section V, which provides a discussion of the key findings. Finally, Section VI concludes the paper by summarizing the major contributions and insights.

II. FRAMEWORK FOR ENERGY CONSUMPTION PREDICTION

This study proposes a hybrid approach combining XGBoost with Exponential Smoothing and explainable AI techniques to forecast energy consumption across various time scales and appliance-level loads. This section, in addition to describing the theories and models used in the proposed methodology, presents the detailed framework and the algorithm implemented for forecasting energy consumption.

A. HOLT-WINTERS EXPONENTIAL SMOOTHING

Holt-Winters Exponential Smoothing [31] is a time series forecasting technique that extends simple exponential smoothing by incorporating both trend and seasonality. It is particularly effective for datasets that exhibit systematic trends along with recurring seasonal patterns. The method can be formulated in either additive or multiplicative form, depending on whether the seasonal variations remain approximately constant or vary proportionally with the series level. In the context of household energy consumption, seasonal variations such as daily load cycles or weekday–weekend effects tend to remain relatively stable in magnitude, irrespective of whether the average demand is high or low. Unlike multiplicative behavior, these patterns do not exhibit proportionally increasing seasonal effects with rising consumption. Hence, the additive form is better suited for modeling the consistent seasonal patterns observed in residential energy demand.

In the additive formulation, the forecasting equation is expressed as:

$$\hat{y}_{t+h} = \ell_t + h \cdot b_t + s_{t+h-m(k+1)} \tag{1}$$

where, ℓ_t is the level (smoothed estimate of the series at time t), b_t is the trend (estimated change in the series), s_t is the seasonal component, m is the number of seasons per year, h is the forecast horizon. The model updates these components recursively using the following equations:

$$\ell_t = \alpha(y_t - s_{t-m}) + (1-\alpha)(\ell_{t-1} + b_{t-1}) \tag{2}$$

$$b_t = \beta(\ell_t - \ell_{t-1}) + (1-\beta)b_{t-1} \tag{3}$$

$$s_t = \gamma(y_t - \ell_t) + (1-\gamma)s_{t-m} \tag{4}$$

where α, β, and γ are smoothing parameters between 0 and 1. The **residual** in Holt-Winters is defined as the difference between the observed value and the fitted value:

$$e_t = y_t - \hat{y}_t \tag{5}$$

This residual reflects the noise or random variation not captured by the level, trend, or seasonal components. Analyzing these residuals is crucial to check whether the model assumptions hold. If the residuals are roughly normally distributed and uncorrelated, it indicates that the model has successfully captured most of the structure in the data. Holt-Winters is widely used in applications like electricity demand forecasting, inventory planning, and weather prediction due to its effectiveness in handling seasonally varying time series.

B. XGBOOST MODEL

XGBoost is a highly effective machine learning model recognized for its superior performance in regression analysis, aiming to develop a robust predictive model [32]. It is an ensemble learning technique based on decision trees. It constructs an additive model in a forward stage-wise manner, where each new tree minimizes the residual errors of the previous ensemble as shown in Figure 1.

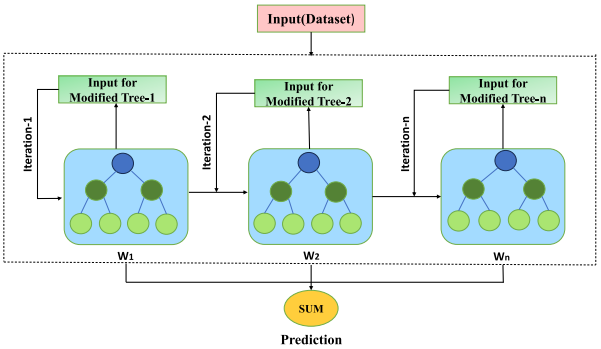


FIGURE 1. Architecture of XGBoost algorithim.

The objective function of XGBoost is defined as:

$$\mathcal{L}(\theta) = \sum_{i=1}^{n} \ell(y_i, \hat{y}_i^{(t)}) + \sum_{k=1}^{t} \Omega(f_k) \tag{6}$$

where:

$$\mathcal{L}(\theta) = \{f_1, f_2, \ldots, f_t\}$$

denotes the set of parameters for all regression trees in the ensemble, - $f_k \in \mathcal{F}$ is the k-th regression tree drawn from the function space $\mathcal{F}$, - $\ell(y_i, \hat{y}_i^{(t)})$ represents the loss function, typically the MSE, - $\Omega(f_k)$ is the regularization term controlling model complexity, e.g., number of leaves and the magnitude of leaf weights. This formulation balances model accuracy and complexity to avoid overfitting during boosting iterations. XGBoost builds the model incrementally:

$$\hat{y}_i^{(t)} = \hat{y}_i^{(t-1)} + f_t(x_i) \tag{7}$$

The regularization term controls model complexity and prevents overfitting:

$$\Omega(f) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2 \tag{8}$$

where, T is the number of leaves in the tree. w_j is the score on the j-th leaf. γ and λ are regularization parameters. To optimize the objective efficiently, a second-order Taylor expansion is used:

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^{n} \left[g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i) \right] + \Omega(f_t) \tag{9}$$

where, $g_i = \partial_{\hat{y}^{(t-1)}} l(y_i, \hat{y}_i^{(t-1)})$ is the first-order gradient. $h_i = \partial^2_{\hat{y}^{(t-1)}} l(y_i, \hat{y}_i^{(t-1)})$ is the second-order derivative of Hessian. Each function f_t corresponds to a regression tree with leaves T and maps an input to a specific leaf:

$$f_t(x) = w_{q(x)}, \quad q : \mathbb{R}^d \to \{1, 2, \ldots, T\} \tag{10}$$

Let $I_j = \{i | q(x_i) = j\}$ be the set of instances of leaf j. The optimal weight w_j^* on each leaf is:

$$w_j^* = -\frac{\sum_{i \in I_j} g_i}{\sum_{i \in I_j} h_i + \lambda} \tag{11}$$

The corresponding optimal value of the objective function becomes:

$$\mathcal{L}_{leaf} = -\frac{1}{2} \sum_{j=1}^{T} \frac{\left(\sum_{i \in I_j} g_i \right)^2}{\sum_{i \in I_j} h_i + \lambda} + \gamma T \tag{12}$$

C. SHAP

SHAP is employed [29]to interpret the predictions generated by the XGBoost model, which is often regarded as a black-box due to its complexity. SHAP values reveal the extent to which each input feature influences the model's output, both at local and global levels. This method improves the transparency of the model, allowing energy analysts to validate predictions based on their domain knowledge. The SHAP values are calculated using the Shapley value formula:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)] \tag{13}$$

where: N is the set of all features, S is a subset of features not containing feature i, $v(S)$ is the value function representing the model output with features in subset S. This formulation has exponential time complexity $O(2^M)$, where M is the number of input features. To address this, the Tree SHAP algorithm computes SHAP values in polynomial time $O(MD^2)$, where D is the maximum tree depth. To improve computational efficiency, a subsampling strategy was adopted: SHAP values were computed on 500 randomly selected test instances using shap. TreeExplainer, ensuring representative but scalable interpretability.

D. LIME

LIME is used with SHAP to provide instance-specific explanations of XGBoost's predictions. By altering input data, LIME trains a more straightforward interpretable model [30] such as linear regression locally around a particular prediction. This allows for a thorough analysis of the effects of minor adjustments to the input variables on the model's output. For example, if a household's predicted energy consumption is surprisingly high, LIME can help identify whether the projection was influenced by increasing use of particular equipment. Both consumers and policymakers seeking to optimize energy use can benefit from such knowledge. The explanation model $\xi(x)$ is defined as the solution to an optimization problem that balances fidelity to the black-box model and interpretability. Specifically, it is formulated as:

$$\xi(x) = \arg\min_{g \in G} \{\mathcal{L}(f, g, \pi_x) + \Omega(g)\} \tag{14}$$

where $\xi(x)$ aims to approximate the black-box model f locally around the instance x using a simpler surrogate model $g \in G$. The objective minimizes a combination of the locally weighted loss $\mathcal{L}(f, g, \pi_x)$, which ensures fidelity to f, and a complexity term $\Omega(g)$, which promotes interpretability. The proximity kernel $\pi_x(z)$ emphasizes samples close to x, enabling the surrogate model to focus on local behavior. Overall, the formulation strikes a balance between predictive accuracy and model simplicity in the generated explanation.

E. PROPOSED METHODOLOGY

The proposed methodology integrates the XGBoost algorithm with XAI techniques to forecast energy consumption across multiple time horizons, hourly to quarterly, and at the level of individual household appliances such as the kitchen, laundry, water heater, and air conditioner. XGBoost is chosen for its exceptional capability to handle structured data and model complex non-linear patterns, making it highly effective for time-series forecasting. To enhance temporal pattern recognition, Holt-winters Exponential Smoothing is applied to extract trend and seasonal components, which are then used as additional features to the model. This hybrid approach improves learning of temporal patterns without stacking models. In addition, to ensure transparency and interpretability, SHAP and LIME are employed to explain

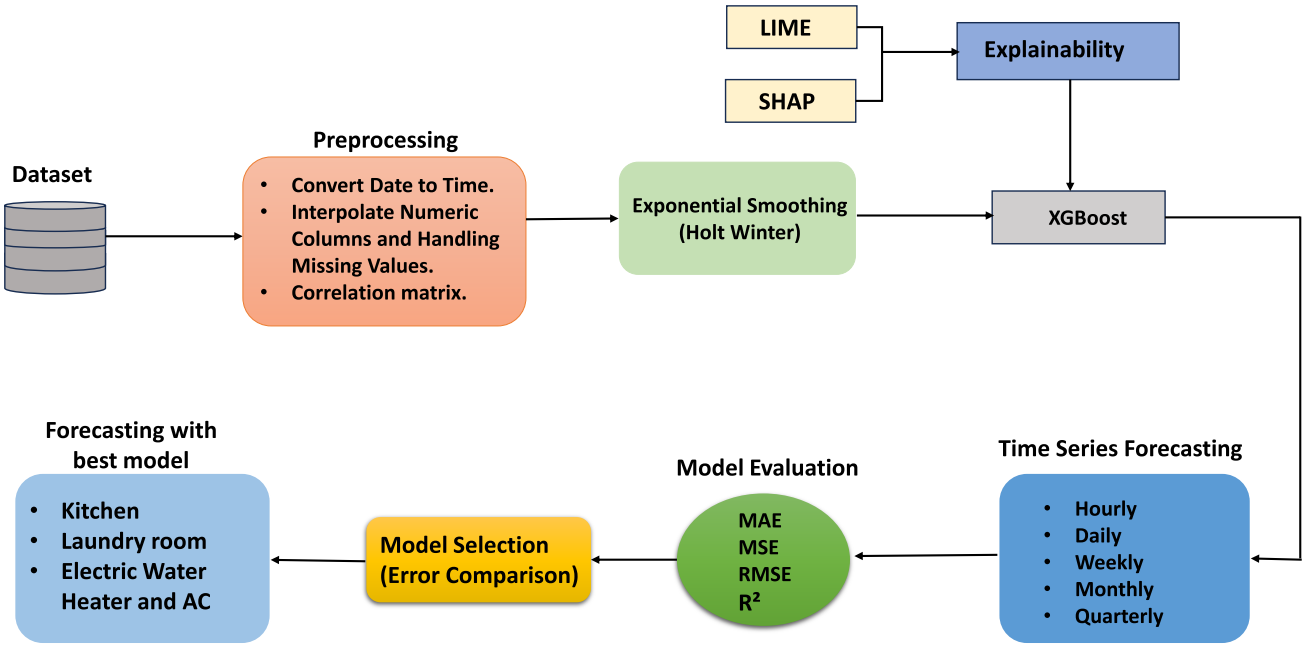


FIGURE 2. Block diagram of the HW+XGBoost+XAI-based energy forecasting framework.

feature influence at both global and local levels. Additionally, PFI and PDP are used to identify and visualize the most impactful variables. The block diagram of the proposed work is depicted in Figure 2.

The algorithm 1 is the training and prediction workflow for the XGBoost model applied to time-series energy consumption forecasting. Initially, raw data is collected and preprocessed to handle missing values, normalize features, and engineer relevant temporal variables. The dataset is then split into training and testing subsets. During training, XGBoost builds an ensemble of decision trees in a sequential manner, where each tree aims to minimize the residual errors of the previous trees using gradient descent optimization. The algorithm constructs histograms for potential feature splits and identifies the best split at each node to grow the tree. After training, the model's explainability is evaluated using techniques like SHAP and LIME to ensure fairness and transparency. If the explanations indicate bias or unfairness, the model is retrained; otherwise, it is deployed. Finally, the trained model is used for prediction on new data, providing interpretable results that help stakeholders understand and trust the forecasted energy patterns.

III. EXPERIMENTATION DETAILS

This section outlines the experimental framework adopted in the study. It first describes the implementation strategy across multiple layers, followed by the dataset characteristics, preprocessing procedures, and decomposition results. Finally, the model training process and selected hyperparameters are presented.

A. IMPLEMENTATION LAYERS

The proposed energy consumption forecasting system is structured into four modular layers, each responsible for a distinct aspect of the pipeline. This architecture enhances scalability, maintainability, and interpretability, making it well-suited for deployment in smart energy systems. The four layers are as follows:

1) DATA LAYER

This layer is responsible for data ingestion, storage, and preprocessing. Raw energy consumption data is collected from smart meters or public datasets and is subjected to rigorous preprocessing. The pipeline is built using Python libraries such as Pandas and NumPy, and data is stored in CSV format. Preprocessing tasks include handling missing values using interpolation, noise reduction via smoothing techniques, and feature extraction such as lag values, and time-based decomposition. Let the raw dataset be denoted as $D = \{(x_t, y_t)\}_{t=1}^{n}$, where x_t represents feature vectors (e.g., sub-metering, voltage) and y_t is the corresponding energy consumption. Preprocessed data $\hat{D}$ is defined as:

$$\hat{D} = \text{Preprocess}(D) = \{(\hat{x}_t, y_t)\}_{t=1}^{n}$$

where $\hat{x}_t$ includes lag features, normalized values, and engineered time-based indicators.

2) MODELING LAYER

This layer trains the forecasting models using the cleaned dataset $\hat{D}$. The XGBoost model minimizes a regularized loss

Algorithm 1 XGBoost Model Training and Prediction

Training Phase:
Acquire raw data: raw_data ← acquire_time_series_data(4)
Preprocess data: preprocessed_data ← preprocess_data (raw_data)
Split data: (train_set, test_set) ← split(preprocessed_data, 0.7)
Train initial model: ml_model ← XGBoost(train_set)
for $t = 1$ to num_trees **do**
  for $i = 1$ to num_features **do**
    for all node in tree_nodes **do**
      usedRows ← get_rows(*node*)
      for $k = 1$ to num_bins **do**
        H ← new Histogram()
        for all $j \in$ usedRows **do**
          bin ← feature_bins[k][j]
          H[bin].sum_y ← H[bin].sum_y + $y[j]$
          H[bin].count ← H[bin].count + 1
        end for
        best_split ← find_best_split(H)
        update_tree(*node*, best_split)
        update_rows(*node*, best_split)
      end for
    end for
  end for
  boosting_update()
  add_tree_to_model()
end for
Explainability and Fairness Evaluation:
explanations ← explain_predictions(*ml_model*, *test_set*)
if explanations_are_unfair(*explanations*) **then**
  ml_model.retrain(*train_set*)
else
  deploy_model(*ml_model*)
end if
Testing Phase:
ml_model ← import_trained_model()
test_set ← acquire_test_data()
predictions ← predict_with_explanations(ml_model, test_set)

function:

$$L(\theta) = \sum_{i=1}^{n} l(y_i, \hat{y}_i) + \sum_{k} \Omega(f_k) \tag{15}$$

where $l(y_i, \hat{y}_i)$ is a loss function such as MSE, and $\Omega(f_k)$ is the complexity penalty for the k-th decision tree. In parallel, time series trends are captured using the Holt-Winters model:

$$\hat{y}_{t+h} = (l_t + hb_t)s_{t+h-m(k+1)} \tag{16}$$

where l_t is the level component, b_t is the trend, s_t is the seasonal factor, and m is the season length. The output of this layer consists of trained model files, performance metrics like MAE, RMSE, and prediction visualizations.

3) EXPLAINABILITY LAYER

To make the machine learning model interpretable, this layer integrates XAI techniques such as SHAP and LIME. SHAP provides both global and local interpretability by computing the contribution of each feature to the model's output. LIME builds interpretable surrogate models locally around each prediction to highlight which features influenced the decision. This layer generates SHAP summary plots, force plots, LIME explanations, and permutation feature importance charts using matplotlib and seaborn.

4) INTERFACE AND DEPLOYMENT LAYER

This final layer focuses on making the forecasting and explanation results accessible and actionable. It includes a real-time dashboard built with Streamlit to visualize predictions, SHAP values, and time series plots. REST APIs are deployed using Flask for model inference. For real-time control, integration with IoT devices such as home automation hubs or grid control panels is facilitated through HTTP protocols. To assess real-time applicability, the system was evaluated for inference latency using a REST API integrated with an IoT simulation. The combined prediction and explanation time–incorporating SHAP and LIME–was approximately 250 milliseconds per instance, demonstrating that the system is viable for near real-time smart energy applications. Further optimizations, such as TreeSHAP caching and surrogate model deployment through model distillation, were also considered to reduce latency in resource-constrained embedded environments. This ensures that energy optimization actions–such as device switching or load balancing–can be taken based on the model's predictions. Overall, this layered architecture not only ensures high predictive accuracy but also enhances interpretability, trust, and real-world applicability of the energy forecasting system.

B. DATASET DETAILS AND PREPROCESSING

This study employs the Individual Household Electric Power Consumption dataset from the UCI Machine Learning Repository [28]. It utilizes six months of one-minute sampled electricity consumption data for a household. This dataset includes a total of 260,640 measurements, offering crucial insights into the understanding of household electricity use. The dataset includes features such as Global-active-power, Voltage, Global-intensity, along with Sub-metering 1 for kitchen data, Sub-metering 2 for laundry data, and Sub-metering 3 for data from the electric water heater and Air Conditioner. The preprocessing techniques applied include handling missing values via interpolation and standardization to normalize features to a unified scale. Date-time parsing is used to extract features such as hour, day of the week, and indicators for weekend/weekday. The correlation matrix in in Figure 3 provides a visual representation of the interdependence among different input features, including sub-metering values, voltage, and global active power. It aids in the identification of multicollinearity and clarifies which features exhibit strong relationships. To effectively model the periodic nature of time-based features such as hour-of-day and day-of-week, cyclical encoding was applied. This approach transforms temporal features using sine and cosine functions to preserve circular continuity. For example, the

hour feature is encoded as:

$$\text{sin_hour} = \sin\left(\frac{2\pi \times \text{hour}}{24}\right) \tag{17}$$

$$\text{cos_hour} = \cos\left(\frac{2\pi \times \text{hour}}{24}\right) \tag{18}$$

This transformation enables the model to correctly interpret that hour 23 and hour 0 are adjacent in time, improving the learning of periodic energy consumption patterns.

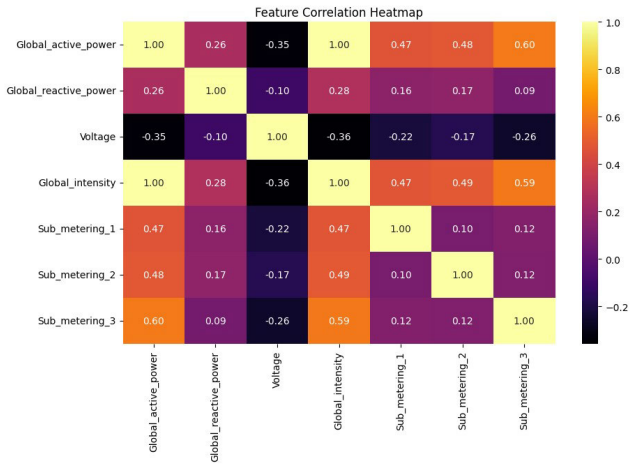


FIGURE 3. Correlation matrix of the features.

The Augmented Dickey-Fuller (ADF) test is performed to check for the stationarity in the dataset. The ADF statistic value of -17.18 and p-value of close to 0, indicates that the time series is stationary. Therefore, no additional differencing is required before applying forecasting models.

C. TREND-SEASONALITY DECOMPOSITION

The time series decomposition of global active power using the Holt-Winters method reveals distinct trend, seasonal, and residual components shown in Figure 4. The observed series exhibits recurring spikes at regular intervals, indicating pronounced cyclical behavior likely driven by daily household energy usage patterns. The extracted trend component shows a gradual increase in average power consumption during the initial period, followed by mild fluctuations and a slight decline toward the end of the observed window, possibly reflecting changes in usage behavior. The seasonal component captures well-defined periodic variations with relatively stable amplitude, confirming the presence of consistent daily or hourly consumption cycles. The residual component, while showing some irregular spikes, remains mostly centered around zero, indicating that the model has effectively captured the underlying structure in the data. Overall, the decomposition highlights the suitability of Holt-Winters smoothing for extracting clear trend and seasonal components, making the data well-suited for forecasting.

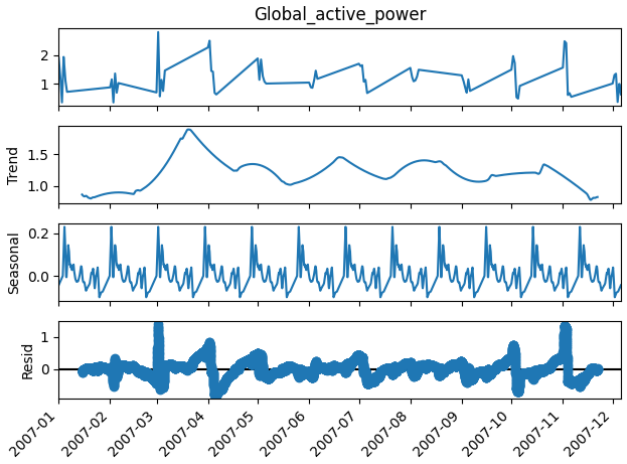


FIGURE 4. Seasonality decomposition for hourly frequency.

TABLE 1. Hyperparameters of the benchmark models.

Model	Parameter	Range	Final
XGBoost	Learning rate	0.01–0.3	0.05
	Max. depth	3–10	6
	Estimators	100–1000	500
	L1 reg (α)	0–1	0
	L2 reg (λ)	0–1	1
LASSO	Alpha	0.0001–1.0	0.01
	Fit intercept	T/F	T
KNN	Neighbors	1–20	5
	Weights	Uniform / Dist.	Dist.
Linear Reg.	Fit intercept	T/F	T
	Normalize	T/F	F
Random Forest	Max features	Auto / Sqrt / Log2	Sqrt
	Estimators	100–1000	300
	Max depth	3–50	None
ARIMA	p (AR)	0–10	2
	d (Int.)	0–2	1
	q (MA)	0–10	2
Prophet	Seas. prior scale	0.01–10	10
	Holidays prior scale	0.01–10	5
	Seasonality mode	Additive / Multiplicative	Additive

D. MODEL TRAINING AND HYPERPARAMETERS

To ensure the model is trained on a diverse range of energy usage patterns, the dataset was split into training and testing sets in an 80-20 ratio, maintaining strict temporal ordering to avoid data leakage. Specifically, the first 80% of chronologically ordered data was used for training, while the most recent 20% was reserved for testing. This setup ensures that the test set strictly follows the training period in time, mimicking real-world forecasting conditions. It is important to note that random shuffling was explicitly avoided, as it would introduce information from the future into the training

set and invalidate the temporal dependencies inherent in time-series forecasting. This forward-chaining method preserves the integrity of the task. The hyperparameters, including the learning rate, number of estimators, and tree depth, are tuned using grid search combined with 5-fold time-aware cross-validation. After the training phase, the model's predictive capabilities are evaluated using standard performance measures such as MAE, MSE, RMSE, and R^2. The key parameters that contribute to the prediction of energy usage are demonstrated through force charts and SHAP summary plots. To improve computational efficiency and ensure the interpretability of SHAP decision plots, a subsampling technique is implemented to reduce processing time.

To evaluate the effectiveness of the proposed model, its performance was benchmarked against five widely used machine learning algorithms: Linear Regression, Least Absolute Shrinkage and Selection Operator (LASSO), K-Nearest Neighbors (KNN), Random Forest, and XGBoost, as well as two timeseries forecasting methods: ARIMA and Prophet. The comparative analysis was conducted across multiple temporal resolutions, including hourly, daily, weekly, monthly, and quarterly forecasting horizons, to assess model robustness and generalizability. To ensure a fair and consistent comparison, the hyperparameters of all baseline models were optimized using grid search combined with cross-validation. The selected configurations for each model are summarized in Table 1.

IV. ENERGY CONSUMPTION FORECASTING RESULTS

This section presents the forecasting results of household energy consumption across multiple temporal resolutions, including hourly, daily, weekly, monthly, and quarterly horizons. A comparative analysis of all methods is then provided to highlight the relative strengths and limitations of each forecasting technique.

A. HOURLY-BASED ENERGY FORECASTING

The predictions made on an hourly basis demonstrate an exceptionally high level of R^2 with 0.999913, with a MAE of 0.000164, MSE of 0.00001, and an RMSE of 0.003447. These minimal error metrics signify the model's robust capability to capture short-term fluctuations in energy consumption. The visualized outcomes validate that the model closely tracks real-time variations. This degree of precision is essential for applications that require precise control, such as real-time grid management or smart appliances. The graph in Figure 5 illustrates highly detailed hourly predictions, reflecting short-term changes in consumption. This degree of specificity is essential for real-time energy management systems, where even slight variations can have considerable consequences.

B. DAILY-BASED ENERGY FORECASTING

On the daily scale, the model maintains a commendable level of R^2 with 0.678933, with an MAE of 0.106480, an MSE of 0.046925, and an RMSE of 0.216622. The model proficiently

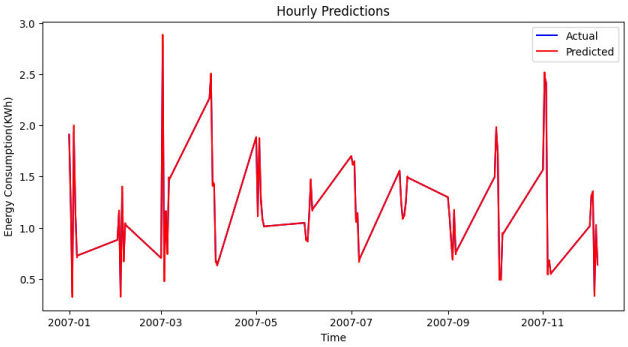


FIGURE 5. Hourly energy consumption forecasting.

captures daily consumption trends and reflects usage patterns, distinguishing between workdays and weekends as illustrated in Figure 6. SHAP visualizations emphasize which features, such as daily sub-metering values, are influential in altering daily demand, enabling stakeholders to make well-informed scheduling or load-balancing decisions. The narrative outlines daily energy forecasts and emphasizes the differences between weekdays and weekends. The accuracy of the model at this level aids in planning energy-heavy activities and enhancing residential energy efficiency.

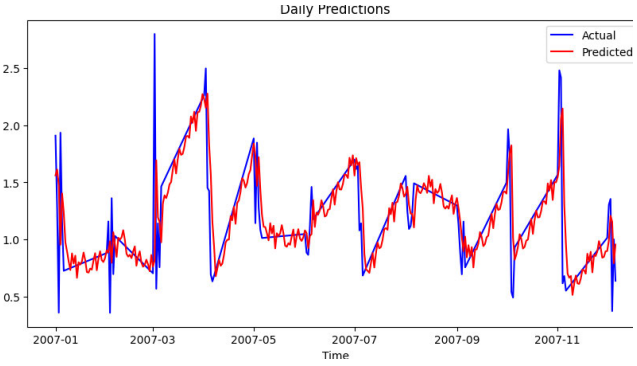


FIGURE 6. Daily based energy consumption forecasting.

C. WEEKLY-BASED ENERGY FORECASTING

In terms of weekly predictions, the model performance remains stable with an R^2 of 0.845577, with an MAE of 0.105337, an MSE of 0.017617, and an RMSE of 0.132730. Weekly forecasts effectively capture cyclical behaviors and periodic peaks, such as increased consumption on weekends. This level of detail is optimal for resource allocation and maintenance planning. The PDP's further illustrate non-linear feature impacts on weekly usage, enhancing interpretability. The weekly prediction plot in Figure 7 highlights the repetitive consumption patterns, including peaks during weekends. This level of temporal detail is well-suited for extensive operational planning and load distribution.

D. MONTHLY BASED ENERGY FORECASTING

The model attains its most favorable results at the monthly level with an R^2 of 0.83953, achieving the lowest error

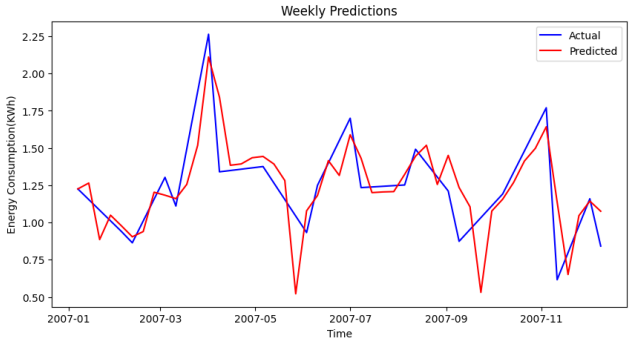


FIGURE 7. Forecasting of energy consumption on a weekly basis.

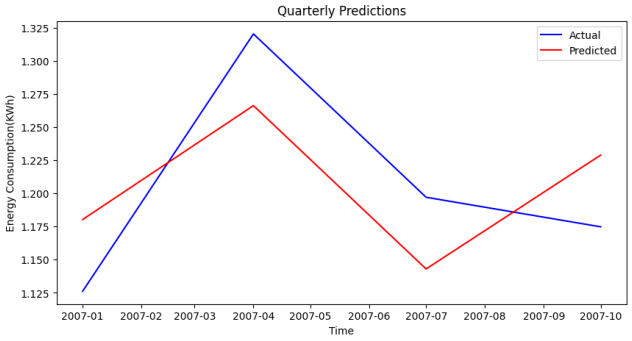


FIGURE 9. Quarterly energy consumption forecasting.

rates–an MAE of 0.138105, an MSE of 0.023276, and an RMSE of 0.152566. The decreased noise at this granularity allows the model to produce smoother forecasts, making it suitable for strategic planning, billing estimates, and long-term capacity management. SHAP and PDP analyses corroborate the significant influence of sub-metering features and stable patterns observed over months. The monthly prediction plot in Figure 8 illustrates the model's consistency on a monthly basis, yielding smooth trends with little noise. These predictions facilitate long-term billing, budgeting, and infrastructure planning.

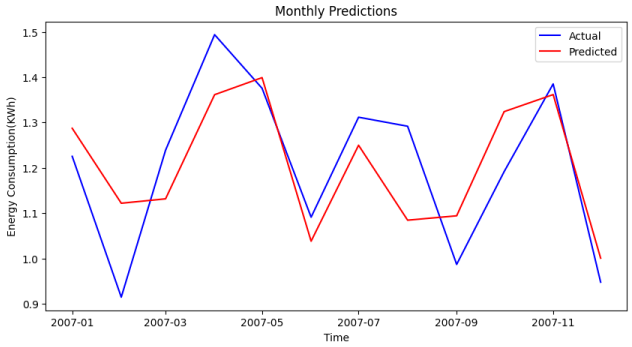


FIGURE 8. Monthly energy consumption forecasting.

E. QUARTERLY BASED ENERGY FORECASTING

Quarterly forecasting provides reliable results with a R^2 of 0.998659, achieving an MAE of 0.006085, an MSE of 0.000047, and an RMSE of 0.006881. This level of detail aids in seasonal planning and budgeting for energy systems. The persistent trends observed throughout the quarters reinforce the reliability of the exponential smoothing technique when combined with XGBoost. It illustrates the influence of broader trends, such as seasonal weather variations or policy shifts, on consumption behaviors. Quarterly forecasts in Figure 9 show seasonal trends and wider consumption patterns.

F. COMPARATIVE ANALYSIS

Seven baseline models, including Linear Regression, LASSO, Random Forest, KNN, XGBoost, ARIMA and Prophet are evaluated across various time granularities (hourly, daily, weekly, monthly, and quarterly) utilizing four essential performance metrics: MAE, MSE, RMSE, and R2, alongside the proposed HW+XGB+XAI. consistently demonstrates superior performance, achieving the lowest error rates and the highest R^2 values across all time granularities, with particularly remarkable results at the hourly level (MAE = 0.000164, R^2 = 0.999913). Although traditional XGBoost also shows commendable performance, its accuracy diminishes at larger time scales. Linear Regression and KNN reveal comparatively higher error rates and lower R^2 values, particularly in the weekly and monthly assessments, suggesting a limited ability to capture intricate time-series patterns. Lasso Regression exhibit moderate performance. ARIMA performs better for the short-term forecasting, revealing low error rates and high R^2 values. Prophet has performed well in long-term forecasting with good error an R^2 values. In summary, the findings highlight the advantages of advanced proposed model for accurate energy forecasting across both short- and long-term horizons, as detailed in Table 2, with error comparisons visualized using RMSE in Figure 10 and R^2 values in Figure 11.

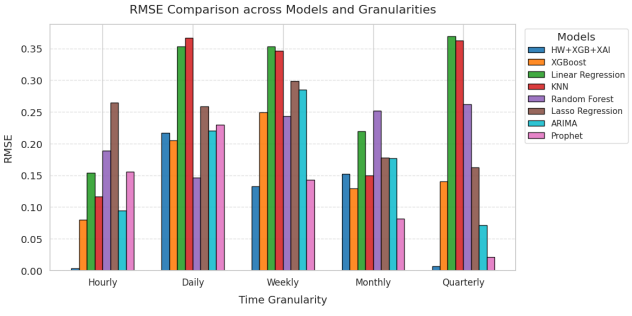


FIGURE 10. Comparison of RMSE for all models.

V. DISCUSSION

The analysis demonstrates that the proposed method significantly enhances both the precision and interpretability of energy consumption predictions. Appliance-wise energy usage, such as that of the kitchen, laundry, and

TABLE 2. Performance metrics of each model for multiple time horizons.

Forecast granularity	Model	Performance Metrics			
		MAE	MSE	RMSE	R^2
Hourly	Linear Regression	0.245910	0.237520	0.154117	0.898765
	LASSO	0.216281	0.070002	0.264579	0.510963
	Random Forest	0.054733	0.809831	0.189153	0.730911
	KNN	0.287049	0.258267	0.116327	0.872514
	XGBoost	0.028897	0.006401	0.080004	0.955020
	ARIMA	0.045687	0.008891	0.094291	0.937936
	Prophet	0.103975	0.024121	0.155309	0.831640
	Proposed (HW+XGB+XAI)	0.000164	0.00001	0.003447	0.999913
Daily	Linear Regression	0.162381	0.124724	0.353162	0.571052
	LASSO	0.293212	0.135884	0.258625	0.530425
	Random Forest	0.096216	0.034709	0.146304	0.835812
	KNN	0.193670	0.934591	0.366866	0.513384
	XGBoost	0.104301	0.042065	0.205096	0.801018
	ARIMA	0.132814	0.048729	0.220746	0.691562
	Prophet	0.155852	0.052953	0.230116	0.665810
	Proposed (HW+XGB+XAI)	0.106480	0.046925	0.216622	0.678933
Weekly	Linear Regression	0.224569	0.124913	0.353431	0.525318
	LASSO	0.219586	0.089405	0.299007	0.635647
	Random Forest	0.180634	0.059087	0.243078	0.576695
	KNN	0.244705	0.119771	0.346079	0.565444
	XGBoost	0.166331	0.062254	0.249508	0.554006
	ARIMA	0.211599	0.081255	0.285053	0.619912
	Prophet	0.106336	0.020439	0.142966	0.758601
	Proposed (HW+XGB+XAI)	0.105337	0.017617	0.132730	0.845577
Monthly	Linear Regression	0.169970	0.048157	0.219448	0.673002
	LASSO	0.156746	0.031681	0.177991	0.743928
	Random Forest	0.094206	0.014855	0.251880	0.551455
	KNN	0.131477	0.022471	0.149904	0.810579
	XGBoost	0.114869	0.016693	0.129201	0.870057
	ARIMA	0.148505	0.031414	0.177240	0.824785
	Prophet	0.071602	0.006719	0.081971	0.786106
	Proposed (HW+XGB+XAI)	0.138105	0.023276	0.152566	0.839543
Quarterly	Linear Regression	0.887627	0.609904	0.369519	0.554283
	LASSO	0.149634	0.041263	0.162354	0.757821
	Random Forest	0.100702	0.014729	0.262362	0.638356
	KNN	0.600702	0.914729	0.362362	0.537347
	XGBoost	0.120265	0.019748	0.140527	0.823179
	ARIMA	0.057803	0.005111	0.071490	0.918126
	Prophet	0.016779	0.000450	0.021203	0.912038
	Proposed (HW+XGB+XAI)	0.006085	0.000047	0.006881	0.998659

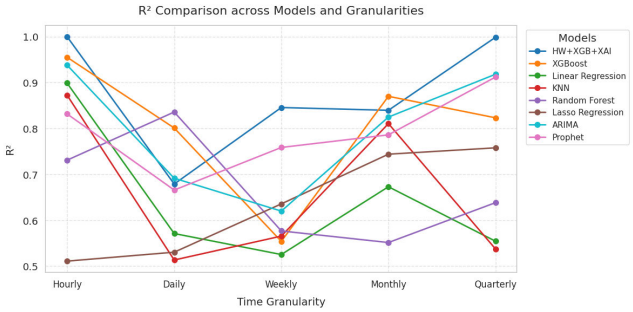


FIGURE 11. Comparison of R^2 for all models.

water heater&AC was forecasted using the proposed HW+XGB+XAI model, as illustrated in Figure 14. The corresponding performance metrics for each appliance are summarized in Table 3, highlighting the model's effectiveness across varying usage patterns.

A. KITCHEN ENERGY FORECASTING

The prediction of energy consumption specific to the kitchen area offers vital insights into the usage patterns of typical kitchen appliances, including refrigerators, microwaves, electric stoves, and dishwashers. These appliances frequently operate on different schedules and power levels some run continuously, like refrigerators, while others function intermittently, such as microwaves and electric cookers. By isolating the energy profile of the kitchen, the model enables energy analysts and homeowners to identify which devices have the greatest impact on electricity costs and whether

TABLE 3. Performance metrics for appliance-level energy consumption forecasting.

S.No	Rooms	MAE	MSE	RMSE	R^2
1	Kitchen	3.5819×10^{-7}	6.5571×10^{-10}	2.6×10^{-5}	0.9598
2	Laundry	2.6815×10^{-7}	8.4345×10^{-10}	2.9×10^{-5}	0.9725
3	Water Heater & AC	1.5656×10^{-6}	9.4520×10^{-9}	9.7×10^{-5}	0.9139
4	Total Active Power	2.0823×10^{-7}	1.7918×10^{-10}	1.3×10^{-5}	0.9955

specific usage trends are linked to inefficient practices. For instance, persistent energy spikes during non-meal times may suggest standby losses or faulty appliances. Furthermore, predicting consumption in this sector assists in developing automation strategies, such as deactivating heavy appliances when they are not in use or scheduling activities such as dishwashing during off-peak periods. These predictions can contribute to lowering both expenses and grid demand during peak times. Therefore, monitoring kitchen-specific energy usage through submetering-1 is crucial for effective energy optimization. The kitchen energy consumption prediction is shown in Figure 12.

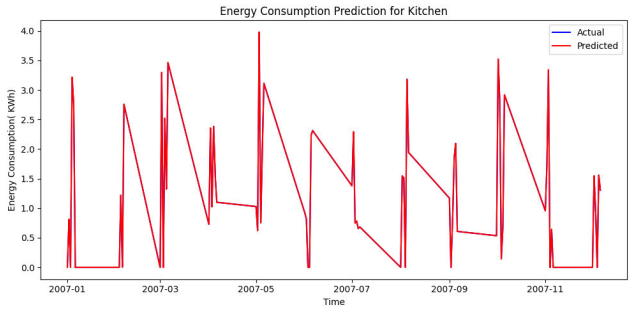


FIGURE 12. Kitchen energy consumption forecasting.

B. LAUNDRY ENERGY PREDICTION

The energy consumption trends observed in the laundry is important, as they typically involve high energy appliances like washing machines, dryers, and ironing devices. These appliances frequently draw significant amounts of current in brief intervals, particularly during spin or heating cycles. It is crucial to analyze consumption in this area by measuring from submetering-2, as these loads are usually flexible and can be scheduled during off-peak hours to alleviate pressure on the energy grid and reduce electricity costs. For residential users, comprehending the frequency and timing of laundry activities can help determine if laundry is being performed inefficiently, for instance, an under-loaded washing cycle. From the perspective of a utility provider, aggregating such usage patterns from various households can support demand-side management strategies. Additionally, forecasting within the laundry sector facilitates integration with smart systems that can suggest optimal operation times or alert users to consumption irregularities such as increased energy usage due to obstructed dryer vents. These features render the laundry a significant candidate for both optimization and safety-related analysis, as illustrated in Figure 13.

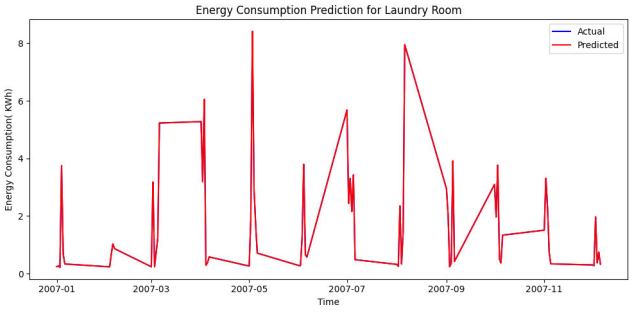


FIGURE 13. Laundry energy consumption forecasting.

C. ELECTRIC WATER HEATER AND AIR CONDITIONER ENERGY FORECASTING

Estimated energy usage associated with electric water heaters and air conditioning units, measured by submetering-3, is of significant importance due to the persistently high energy requirements of these devices. Water heaters generally function in cycles, preserving water temperature even when not actively utilized, which results in a parasitic load that frequently goes unnoticed. Likewise, air conditioners draw considerable power depending on the set temperature, room insulation, external weather conditions, and user habits. The forecasting model's capability to accurately predict usage trends in this context enables informed choices regarding thermal comfort with cost efficiency. By analyzing these trends, energy managers can implement load-shedding or adaptive control measures, such as pre-cooling during periods of low tariffs or employing energy storage systems to accommodate heating demands. Moreover, the insights gained from these forecasts can aid in maintenance planning—unusual spikes in consumption may signal inefficiencies like refrigerant leaks or deterioration of heating elements. Additionally, on a broader scale, predicting the loads of air conditioning and water heating systems is crucial for planning for peak demand, particularly in regions with extreme temperatures. Therefore, this data highlights the significance of precise forecasting in minimizing operational expenses, enhancing comfort, and promoting energy sustainability, The electric water heater and air conditioner energy consumption prediction is shown in Figure 15.

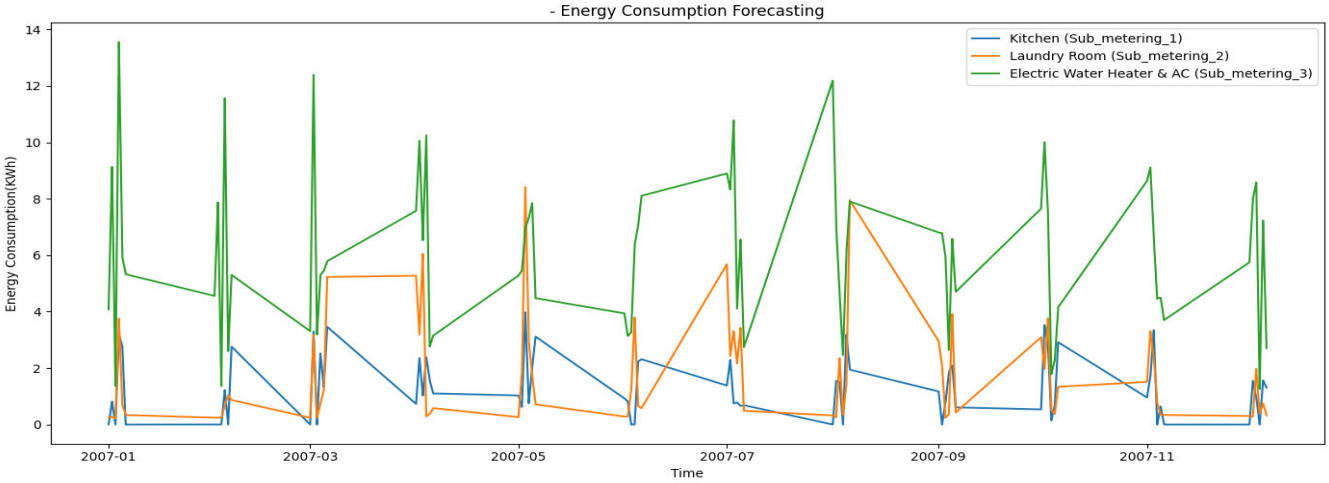


FIGURE 14. Forecasting of household energy consumption using the proposed model.

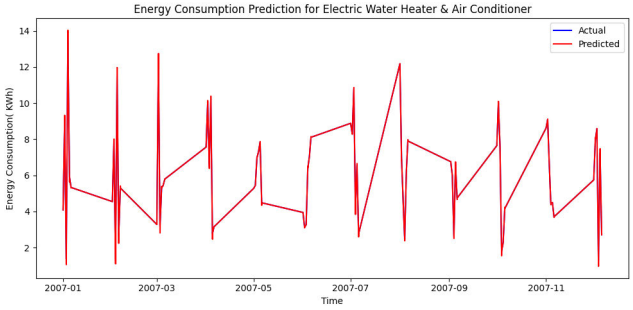


FIGURE 15. Electric water heater and air conditioner energy consumption forecasting.

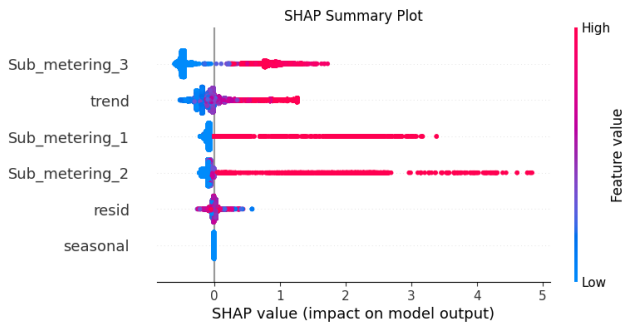


FIGURE 16. SHAP value impact on model prediction.

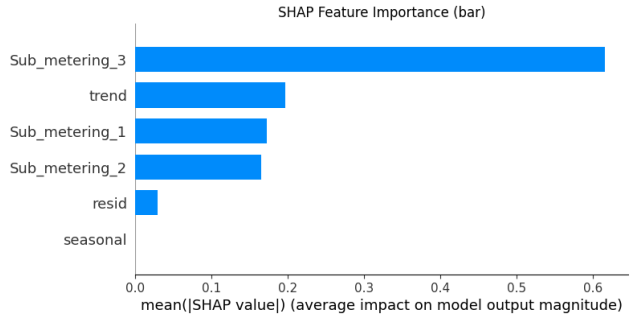


FIGURE 17. Average SHAP value impact on model output.

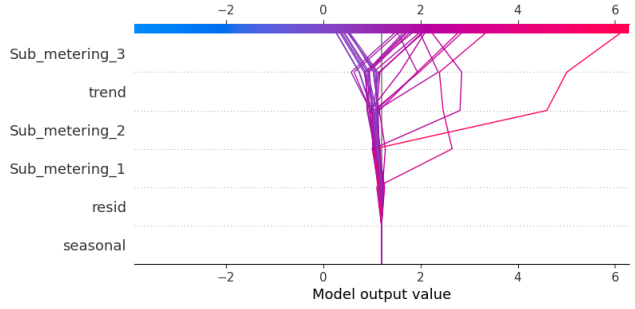


FIGURE 18. SHAP summary plot for feature contribution.

D. EXPLAINABILITY WITH SHAP FOR MODEL INTERPRETATION

This SHAP value plot demonstrates the role of each feature in a specific prediction. Each dot signifies a SHAP value corresponding to a feature for a single instance, indicating the extent to which the feature affected the prediction result. It highlights both the direction and magnitude of the influence that individual features exert on the model's output, as illustrated in Figure 16 and average impact on a model is shown in Figure 17.

This SHAP summary plot offers a comprehensive interpretation of the model by illustrating the average absolute SHAP values for each feature. Features such as sub-metering 3 and global active power are positioned at the top, signifying their significant impact on energy consumption predictions, as depicted in Figure 18 and also SHAP values variance of features on model output has shown in Table 4.

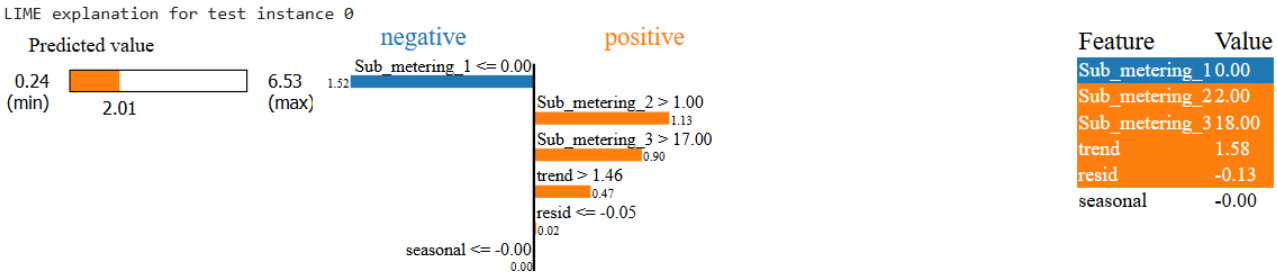


FIGURE 19. LIME explainer plot for a sample instance.

TABLE 4. SHAP variance of features on model output.

Feature	SHAP Variance
Sub_metering_3	0.426118
Sub_metering_2	0.193634
Sub_metering_1	0.191464
trend	0.101936
resid	0.001950
seasonal	0.000000

E. INSTANCE-BASED EXPLAINABILITY USING LIME

LIME outputs used to interpret the predictions made by a machine learning model, particularly focusing on the contribution of different features to the predicted global active power. The Figure 20 shows LIME stability analysis across multiple runs, measuring how consistent the model explanations are. It displays the thresholds for each feature e.g., Submetering2 > 1.00, trend > 1.46 along with the delta values, indicating variation in feature contributions across runs lower delta implies more stable interpretations. The Figure 19 provides a LIME explanation for a specific test instance, showing which features positively (orange) and negatively (blue) contributed to the prediction. For instance, Submetering2, Submetering3, and trend increased the predicted value, while Submetering1 had a strong negative influence. The feature values used in that prediction are also listed on the right. Together, these visuals demonstrate both the local interpretability and stability of the model's behavior.

F. FEATURE IMPORTANCE ANALYSIS AND PARTIAL DEPENDENCE PLOTS

A Permutation Feature Importance study is performed to assess the influence of randomly rearranging each feature on model performance in order to gain a deeper understanding of the significance of individual features. Future model improvements will be guided by features that have been recognized as major contributors to energy consumption and have high significance rankings. PDPs are also produced to show non-linear correlations between predictions and input variables. PDPs offer a natural understanding of the behaviour of the model by showing how altering one feature while maintaining the rest unchanged impacts energy consumption predictions. The narrative examines the interdependence of features through the use of SHAP dependence plots. This approach facilitates the comprehension of how the impact of one feature varies in relation to another, thereby assisting analysts in interpreting non-linear dependencies, as illustrated in Figure 21. The narrative illustrates the impact of shuffling the values of each feature on the performance of

```
LIME Stability Run 1 vs 2:
Sub_metering_1 <= 0.00 <= 0.00: Δ = 0.0359
Sub_metering_2 > 1.00 <= 0.00: Δ = 0.1564
Sub_metering_3 > 17.00 <= 0.00: Δ = 0.0028
trend > 1.46 <= 0.00: Δ = 0.0421
resid <= -0.05 <= 0.00: Δ = 0.0139
seasonal <= -0.00 <= 0.00: Δ = 0.0000

LIME Stability Run 2 vs 3:
Sub_metering_1 <= 0.00 <= 0.00: Δ = 0.0566
Sub_metering_2 > 1.00 <= 0.00: Δ = 0.0749
Sub_metering_3 > 17.00 <= 0.00: Δ = 0.0188
trend > 1.46 <= 0.00: Δ = 0.0367
resid <= -0.05 <= 0.00: Δ = 0.0239
seasonal <= -0.00 <= 0.00: Δ = 0.0000
```

FIGURE 20. LIME Stability score.

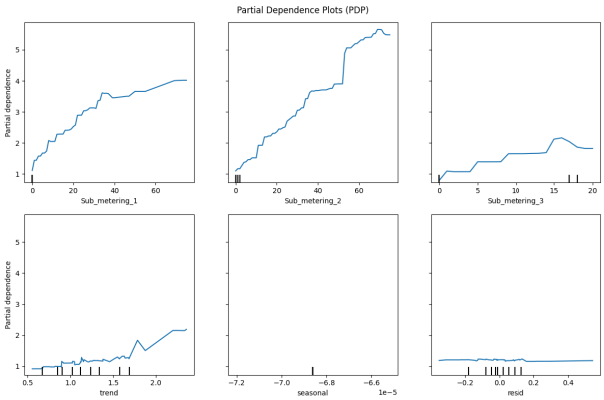


FIGURE 21. Non-linear feature relationships in partial dependence Plots.

the model, ranking the features according to their predictive capabilities. A notable decrease in accuracy for specific features such as sub-metering 3 underscores their essential significance, as depicted in Figure 22.

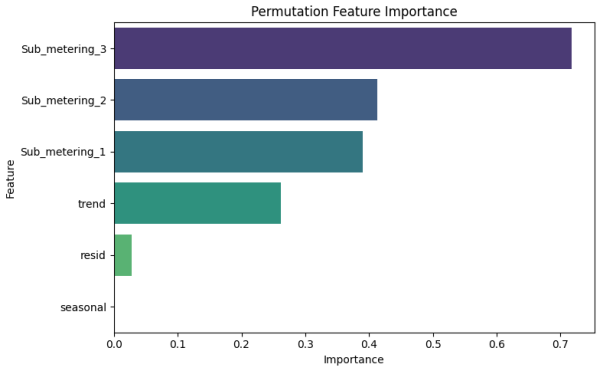


FIGURE 22. Permutation feature importance for energy consumption prediction.

VI. CONCLUSION

This study demonstrates the effectiveness of integrating Holt-Winters decomposition with XGBoost and XAI techniques for energy consumption forecasting. By leveraging the combined strengths of Holt-Winters, XGBoost, and interpretability frameworks such as SHAP and LIME, the proposed model effectively addresses challenges associated with time-series forecasting. The framework achieved high predictive accuracy, with an R^2 of 0.9999 for hourly forecasting and 0.9986 for quarterly forecasting, along with consistently low RMSE values. Beyond accuracy, the HW+XGB+XAI model enhances model transparency by identifying the contributions of key features through SHAP, LIME, PFI, and PDP analyses. By delivering both precise forecasts across multiple time horizons and transparent insights into model behavior, the proposed approach bridges the gap between performance and interpretability. These outcomes enable stakeholders to make informed decisions, thereby supporting more efficient and sustainable energy management systems.

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